



Programme: Data Science and Business Analytics

CALCULUS

Lecture 2 (14.09.2026)

Sequences. Infinitesimal and infinitely large sequences and their properties.

1. Main definitions.

Suppose we are given a rule that assigns an object (a number) a_n to each natural $n \in \mathbb{N}$. Then we write this rule explicitly in the form

$$\{a_1, a_2, \dots, a_n, \dots\}$$

or, more briefly, in the form $\{a_n\}$. Further we shall say that we are given *the sequence* $\{a_n\}$.

Definition 1. A sequence $\{a_n\}$ of real numbers is said to be *bounded above* if there is a number K such that $a_n \leq K$ for all $n \in \mathbb{N}$. A sequence $\{a_n\}$ is *bounded below* if there is a number L such that $a_n \geq L$ for $n \in \mathbb{N}$. If $\{a_n\}$ is bounded above and below, then $\{a_n\}$ is said to be *bounded*.

Definition 2. A sequence $\{a_n\}$ is said to be *unbounded* if for any number $A > 0$ there is such $n \in \mathbb{N}$ (or, there is such an element a_n of the sequence) that $|a_n| > A$.

Definition 3. A sequence $\{a_n\}$ is said to be *infinitely large*, that is,

$$\lim_{n \rightarrow \infty} a_n = \infty$$

if for any number $A > 0$ there is such $N \in \mathbb{N}$ that for any $n > N$: $|a_n| > A$.

Remark 1. It is easy to see that any infinitely large sequence is unbounded. On the other hand, unbounded sequence is not always infinitely large. For instance, the unbounded sequence $\{1, 2, 1, 3, 1, 4, \dots, 1, n, \dots\}$ is unbounded but it is not infinitely large as all odd terms of this sequence are equal to 1.

Definition 4. A sequence $\{\alpha_n\}$ is said to be *infinitesimal*, or *infinitely small*, that is,

$$\lim_{n \rightarrow \infty} \alpha_n = 0$$

if for any number $\varepsilon > 0$ there is such $N \in \mathbb{N}$ that for any $n > N$: $|\alpha_n| < \varepsilon$: for

$$\forall \varepsilon > 0 \exists N \forall n > N : |\alpha_n| < \varepsilon.$$

Example 1. Let us consider the sequence $\{\alpha_n\} = \left\{\frac{1}{n}\right\}$. For this sequence to satisfy the condition

$$\frac{1}{n} < \varepsilon \quad (1)$$

n should be more than $\frac{1}{\varepsilon}$. So, if $N = \left[\frac{1}{\varepsilon}\right] + 1$ where $[x]$ is the integer part of the number x , we have (1).

Definition 5. A number a is said to be a limit of the sequence $\{a_n\}$,

$$\lim_{n \rightarrow \infty} a_n = a \quad (2)$$

if for any number $\varepsilon > 0$ there is such $N \in \mathbb{N}$ that for any $n > N$: $|a_n - a| < \varepsilon$:

$$\forall \varepsilon > 0 \quad \exists N \quad \forall n > N : |a_n - a| < \varepsilon.$$

If a finite limit of a sequence $\{a_n\}$ exists then $\{a_n\}$ is said to converge (the sequence is convergent).

If the limit (2) doesn't exist then $\{x_n\}$ is said to diverge (the sequence is divergent).

Remark 2. Let us remark that if $\alpha_n = x_n - a$ then $\{\alpha_n\}$ is infinitesimal as it follows from the Definition 5. So, any convergent sequence $\{x_n\}$ such that $\lim_{n \rightarrow \infty} x_n = a$ can be presented in the form

$$x_n = \alpha_n + a \quad (3)$$

where $\{\alpha_n\}$ is infinitesimal.

2. Properties of infinitesimal and infinitely large sequences.

Theorem 1. If $\{\alpha_n\}$ and $\{\beta_n\}$ are infinitesimal then their sum (difference) $\{\alpha_n \pm \beta_n\}$ and product $\{\alpha_n \cdot \beta_n\}$ are also infinitesimal sequences.

Proof. If sequences $\{\alpha_n\}$ and $\{\beta_n\}$ are infinitesimal, by definition,

$$\forall \varepsilon > 0 \exists N_1 \forall n > N_1 : |\alpha_n| < \varepsilon/2 \quad (4)$$

and

$$\forall \varepsilon > 0 \exists N_2 \forall n > N_2 : |\beta_n| < \varepsilon/2. \quad (5)$$

Assume that $N = \max\{N_1, N_2\}$. Then $\forall n > N$

$$|\alpha_n \pm \beta_n| \leq |\alpha_n| + |\beta_n| < \varepsilon/2 + \varepsilon/2 = \varepsilon.$$

We get that the sequence $\{\alpha_n \pm \beta_n\}$ is infinitesimal.

To prove the second part of the theorem let us remark that, from (5) it follows that there exists such N_3 that for any $n > N_3$ the condition $|\beta_n| < 1$ is valid. So, if we put $N = \max\{N_1, N_3\}$ we obtain that $\forall n > N$

$$|\alpha_n \cdot \beta_n| = |\alpha_n| \cdot |\beta_n| < \frac{\varepsilon}{2} \cdot 1 < \varepsilon$$

So, $\{\alpha_n \cdot \beta_n\}$ is also infinitesimal. □

Theorem 2. If $\{\alpha_n\}$ is infinitesimal and $\{x_n\}$ is a bounded sequence then their product $\{\alpha_n \cdot x_n\}$ is also infinitesimal sequence.

Proof. If $\{x_n\}$ is bounded, by definition, there exists a real number $A > 0$ such that for any $n \in \mathbb{N}$ $|x_n| < A$. If $\{\alpha_n\}$ is infinitesimal then

$$\forall \varepsilon > 0 \exists N \forall n > N : |\alpha_n| < \frac{\varepsilon}{A}$$

So, $\forall n > N$

$$|\alpha_n \cdot x_n| = |\alpha_n| \cdot |x_n| < \frac{\varepsilon}{A} \cdot A < \varepsilon,$$

and $\{\alpha_n \cdot x_n\}$ is also infinitesimal. □

Theorem 3. If $\{x_n\}$ is infinitely large and $x_n \neq 0$ for any $n \in \mathbb{N}$ then the sequence $\{\alpha_n = \frac{1}{x_n}\}$ is infinitesimal. Vice versa, if $\{\alpha_n\}$ is infinitesimal and $\alpha_n \neq 0$ for any $n \in \mathbb{N}$ then the sequence $\{x_n = \frac{1}{\alpha_n}\}$ is infinitely large.

Proof. Suppose that $\{x_n\}$ is infinitely large, that is, $\forall A > 0 \exists N \in \mathbb{N} \forall n > N : |x_n| > A$. Put $\varepsilon = 1/A$ then $\forall n > N$

$$|\alpha_n| = \left| \frac{1}{x_n} \right| < \frac{1}{A} = \varepsilon,$$

so that $\{\alpha_n\}$ is infinitesimal. The second part of the theorem can be proved analogously. □